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Substrate Stability Under Persistent Disagreement: Structural Constraints on Individuation in Neutral Ontological Substrates --Manuscript Draft--

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Abstract:	Modern data systems increasingly operate under conditions of persistent legal, political, and analytic disagreement. In systems that adopt a neutral substrate, shared interpretation cannot be assumed, and representations must preserve stable reference across incompatible extensions. This paper examines the structural consequences of that architectural choice. Treating interpretive neutrality and extension-stable identity as explicit design constraints, the analysis asks what minimal ontological structure is required for entities and their relations to remain referable and comparable under persistent disagreement. Minimality is defined structurally: reductions are admissible only if they do not reintroduce identity-relevant distinctions through hidden roles, flags, or contextual predicates. A conditional lower bound is established: any ontology intended to function as a neutral substrate under persistent disagreement must realize at least six distinct identity-and-persistence regimes. A construction with exactly six regimes satisfies the stated requirements without embedding causal or normative commitments in the substrate. Together these results establish a tight structural bound on entity individuation in neutral substrates. Preview of Main Result. Under explicit neutrality and stability constraints, attempts to simplify ontological structure by collapsing distinctions reintroduce those distinctions as hidden interpretive mechanisms. Excluding such hidden regimes implies six distinct identity-and-persistence regimes required for accountability under persistent disagreement: fewer do not satisfy the stated requirements, and additional regimes introduce no further structural capacity. Informally. Given a neutral substrate capable of supporting persistent interpretive disagreement, entities and relations must remain stably referable across incompatible interpretive frameworks. This requirement constrains how entities may be individuated. Attempts to simplify the individuation structure by collapsing categories reintroduce those distinctions as hidden interpretive mechanisms. Conversely, a construction with six identity-and-persistence regimes provides sufficient structure to preserve stable

	reference without embedding causal or normative commitments in the substrate. Together these results establish a tight structural bound of six regimes.
Response to Reviewers:	Dear Editor and Reviewers, Many thanks to the reviewers for their detailed engagement. The paper has been substantially revised to clarify its contribution, strengthen the derivation, and improve presentation. The revision makes the central contribution explicit: the paper does not propose new ontological categories; it derives and realizes a minimal identity-and-persistence regime structure under neutrality and extension-invariant identity. The result establishes both a lower bound (necessity) and a construction achieving that bound (sufficiency), and is therefore a conditional structural constraint on neutral substrate design. The subtitle has been updated to better reflect the scope (Substrate Stability Under Persistent Disagreement: Structural Constraints on Individuation in Neutral Ontological Substrates). The main revisions are: Section 1 (Introduction: Structure Under Disagreement) - Strengthened the motivation through diagnostic cases (regulatory reporting, legislative history, provenance), and clarified their role in motivating the requirements. - Clarified the result as a conditional constraint under neutrality and stability across incompatible frameworks, and more clearly introduced the notion of hidden regimes. Section 2 (Assumptions and Design Constraints) - Made all assumptions explicit, including neutrality as a design constraint and extension without coordination. - Clarified persistent disagreement and stability, emphasizing regime-level identity-and-persistence invariance. - Strengthened the stable reference requirement and clarified its scope as intentionally strong and conditional. Section 3 (Structural Requirements for Neutral Substrates) - Framed R1-R6 as representational capacities and strengthened their justification with requirement-specific arguments. - Introduced identity-and-persistence regimes as structural modes of individuation determined by invariant identity, persistence, and admissible relations. - Clarified that the requirements constrain but do not determine the regime structure derived in Section 4. Section 4 (Necessity: A Lower Bound on Ontological Structure) - Reworked the argument into an explicit three-step proof (profile extraction, collapse admissibility, failure modes). - Made the derivation of regimes explicit as induced persistence-and-admissibility profiles. - Introduced a formal collapse admissibility criterion and three failure modes (identity instability, category error, hidden regime reintroduction), supported by worked examples and pairwise analysis. - Clarified the derivation of the ENR regime and strengthened the lower-bound lemma. Section 5 (Sufficiency: A Six-Regime Construction) - Provided an explicit construction realizing the six regimes, demonstrating sufficiency and tightness. - Specified a minimal relation set satisfying R1-R6 without hidden discriminators or additional regimes. Section 6 (Interpretation of the Result) - Clarified the result as a conditional structural bound and refined the notion of minimality. - Explained how fixing identity and admissibility at the substrate level enables divergent extension through conservative specialization.

- Positioned the result relative to existing ontologies.

Section 7 (Related Work)

- Repositioned related frameworks to clarify alignment and differences in objectives.

The following provides specific responses to each comment and indicates where associated updates appear in the revision.

Reviewer #1 summarized the paper and noted major problems with this work, especially that it:

- (a) does not motivate itself sufficiently, and
- (b) that it occasionally reads as a self-fulfilling prophecy.

RESPONSE: Agreed. Thank you for these remarks and for recommending specific improvements. The revision addresses each concern as summarized above and described below.

Concern 1. Motivation and real-world relevance remain unclear

The core setting ("persistent disagreement" + "uncoordinated extension" + "neutral substrate") is asserted rather than grounded in concrete deployments. The introduction claims modern systems "increasingly operate" this way, but the paper does not give persuasive applications (e.g., specific accountability workflows, multi-jurisdiction regulatory reporting, cross-institution audit trails, etc.) nor evidence that existing ontology frameworks actually aim for neutrality in this strict sense. Suggestions:

1.1) add 2-3 concrete case studies (even short sketches) and

1.1) RESPONSE: Agreed. Three new diagnostic cases (multi-jurisdiction regulatory reporting, legislative history tracking, and cross-institutional provenance) have been added to the start of Section 1 (Introduction: Structure Under Disagreement). These cases are explicitly reused in Section 3 (Structural Requirements for Neutral Substrates) when motivating requirements R1-R6, and are revisited in Section 6 (Interpretation of the Result) to illustrate the behavior of the resulting regime structure.

1.2) explicitly name candidate frameworks/design patterns that purport to be "neutral substrates"

1.2) RESPONSE: Agreed. The revised Section 7.3 (Related Work / Upper Ontologies) now explicitly situates the problem relative to commonly used ontology frameworks, including BFO and DOLCE, and clarifies the design setting in which neutral substrate deployments arise.

1.3) then show where they fail the stated constraints.

1.3) RESPONSE: Agreed. The revised Section 7.3 (Related Work / Upper Ontologies) now clarifies that the contribution is not a critique of existing frameworks, but identifies points of structural under-constraint that arise when such frameworks are used as neutral substrates under the stated assumptions. In particular, distinctions required for extension-invariant identity and admissibility may be left to implementation or interpretive convention. The analysis makes explicit where additional commitments would be required to satisfy the neutrality and stability constraints.

Concern 2. The Stable Reference Requirement is extremely strong and likely unrealistic

2.1) Section 2.3 effectively requires that entities never need to change "ontological status" (identity conditions / persistence regime) as extensions evolve. But many real scenarios do face Ship-of-Theseus-style identity pressure: the same "thing" is tracked through transformations, splits/merges, and re-identification, e.g., baking ingredients, dow, and bread rolls.

2.1) RESPONSE: Agreed. Section 2.4 (Stable Reference Requirement) now explicitly acknowledges that the stability assumption is strong and that engineering systems may address identity pressure through contextual or negotiated identity schemes.

2.2a) The paper needs to acknowledge that (a) strong reference stability may be unattainable in practice

2.2a) RESPONSE: Agreed. Section 2.4 (Stable Reference Requirement) now explicitly notes that strong reference stability may not be attainable in practical settings, and that systems often instead rely on versioning, contextual identity, or negotiated classification to manage identity pressure. The analysis in this paper is explicitly conditional: it concerns design settings in which stable reference must be preserved independently of interpretive convergence. In settings where identity is managed through versioning or contextual schemes, the stability assumption does not hold, and the lower-bound result does not apply.

2.2b) (2b) engineering solutions often treat identity as versioned, contextual, or negotiated (which the paper would likely classify as "hidden regimes" or reliance on interpretive layers).

2.2b) RESPONSE: Agreed and clarified. Section 2.4 (Stable Reference Requirement) now explains that the analysis concerns design settings in which stable reference must be preserved independently of interpretive convergence and cannot be maintained through contextual or negotiated identity at extension layers. Under the neutrality and extension without coordination assumptions, such mechanisms rely on interpretive agreement and therefore do not satisfy the substrate-level stability requirement. The diagnostic cases illustrate situations in which such reliance is not admissible.

2.2c) Either (2c) defend why this strength is necessary for the target applications, or discuss relaxations and how the bound changes if stability is weakened.

2.2c) RESPONSE: Agreed. The revised Section 6 (Interpretation of the Result) clarifies that the lower-bound result is conditional on the stability assumption. If stability is weakened, the bound derived in the paper no longer applies. Conversely, the strength of the assumption is what enables extension-invariant identity across incompatible frameworks, which is required in the target accountability settings.

Concern 3. Why minimize explicit structure at all?

The optimization criterion (Section 2.4) frames minimality as "minimize explicit identity-and-persistence regimes" and rejects "reductions" that smuggle distinctions via roles/flags/context. Two issues:

3.1) The reason to minimize is not well justified. Many engineering contexts prefer explicit richness (more kinds, more typed relations) to reduce ambiguity and auditing cost, even under disagreement.

3.1) RESPONSE: Agreed. The revision clarifies that the minimality criterion is methodological rather than prescriptive. The analysis asks which identity-and-persistence distinctions are unavoidable under the neutrality and stability constraints. In particular, distinctions are considered eliminable only if they can be removed without reappearing as implicit structure (e.g., through roles, flags, or contextual predicates). Where such reintroduction occurs, the apparent simplification does not reduce structure but relocates it to the extension layer, violating extension-invariant admissibility. Section 6 (Interpretation of the Result) clarifies this criterion and its implications.

3.2) The exclusion of "hidden regimes" makes the objective feel like a scope stipulation rather than a neutral modeling desideratum.

3.2) RESPONSE: Agreed. Section 6.3 (Why Hidden Regimes Are Excluded) presents the exclusion of hidden regimes as a consequence of the neutrality and stability constraints rather than as a modeling stipulation. As stated there, such mechanisms rely on interpretive convergence and are therefore incompatible with the requirement that substrate-level admissibility be preserved under persistent disagreement.

Concern 4. R1-R6 read like asserted scope definitions, not well-supported necessities

4.1) The paper says R1-R6 are "necessary capacities" for accountability substrates

(and "not derived from any particular domain"), but they are presented as if they were close to universal.

4.1) RESPONSE: Agreed. The revised Section 3 (Structural Requirements for Neutral Substrates) makes explicit that R1-R6 describe representational capacities, and that they are requirements for neutrality under the stated assumptions and are not intended as general claims about ontology design.

4.2) This is exactly where the argument feels self-fulfilling: R1-R6 are then mirrored almost one-to-one by K1-K6 (Section 5.1), making the "result" partly look like a restatement of the chosen requirements.

4.2) RESPONSE: Agreed. Section 4 (Necessity: A Lower Bound on Ontological Structure) now makes the derivation explicit by showing how persistence and admissibility profiles induced by the requirements generate the regimes under the collapse admissibility criterion. In particular, the regimes are not introduced as restatements of R1-R6, but arise as the minimal construction under which admissible relations and persistence conditions remain invariant under extension. Attempts to collapse regimes either violate extension-invariant identity, produce category error, or reintroduce hidden discriminators. The result derives structural constraints on individuation from the stated assumptions rather than restating requirements.

4.3) Concrete concern example: R4 (Time-indexed occurrence) is asserted as required. But it's not obvious this is necessary for every accountability substrate. Many accountability claims can be tracked via records, states, or obligations without committing the substrate to a specific occurrence ontology—especially if "occurrence" individuation itself is contested.

4.3) RESPONSE: Agreed. Section 3 (Structural Requirements for Neutral Substrates) now clarifies that R4 is not introduced as a modeling preference for events, but as a representational requirement forced by the need to anchor temporal realization independently of interpretive commitments. Representations based solely on records, states, or obligations can describe occurrences, but do not provide a substrate-level referent for them. As a result, participation and provenance relations must either attach to descriptive artifacts or be reconstructed from them. Under persistent disagreement, such representations make identity and admissibility dependent on interpretive reconstruction, violating extension-invariant reference as required by R1. Attempts to enforce stable anchoring within such representations either introduce identity conditions through relations or reintroduce a distinct regime implicitly (e.g., via special record types), thereby violating the admissibility criterion or the exclusion of hidden regimes. Section 3 makes this explicit through requirement-specific arguments and examples, showing that without a regime that fixes temporal realization, accountability comparisons cannot be maintained under incompatible extensions.

4.4) Suggestion: For each of R2-R6, give either (i) a brief argument schema ("without Rk, accountability comparisons fail because ..."), or (ii) domain-driven evidence/examples showing that capacity is required in the target applications.

4.4) RESPONSE: Agreed. Section 3 (Structural Requirements for Neutral Substrates) now provides, for each of R2-R6, a brief argument schema showing that the corresponding capacity is necessary under the neutrality and extension-stability constraints. In each case, eliminating the capacity does not remove the underlying distinction, but forces it to be represented either implicitly through roles, flags, or contextual predicates, or through interpretive reconstruction from descriptive artifacts. Under persistent disagreement, these alternatives make identity or admissibility dependent on interpretive convergence, violating extension-invariant reference as required by R1. Section 3 makes these failure modes explicit, showing that each requirement corresponds to a representational distinction that cannot be eliminated without either embedding causal or normative commitments in the substrate or reintroducing hidden regimes under the stated constraints.

Concern 5. Verbosity and repetition (and citation style)

5.1) There is substantial repetition and rephrasing, e.g., Sections 2.3 vs .3.1.

5.1) RESPONSE: Agreed. The revision reduces repetition across sections, in particular by consolidating overlapping statements about stability and neutrality between Section 2.4 (Stable Reference Requirement) and Section 3.1 (R1: Stable Identity and Persistence (Global Constraint)), and streamlines the presentation throughout.

5.2) Also the citation format "Author [Year]" throughout hurts readability in dense passages (same issue as the other paper).

5.2) RESPONSE: Agreed. Citation format has been revised to standard (author year) style throughout.

Concern 6. Section 4.2 needs at least one worked "collapse error" example

6.1) Section 4.2 outlines three failure modes of collapsing regimes—identity instability, category error, hidden-regime reintroduction—but does not provide a crisp, worked example.

6.1) RESPONSE: Agreed. Section 4.4 (Pairwise Collapse Analysis) now includes explicit collapse analyses with worked examples, including OBL/ENR, OBL/NOR, NOR/CTX, NOR/REC, and OCC/ENR, together with a summary table and an exhaustive characterization of failure modes. In addition, Appendix A. Complete Pairwise Collapse Enumeration provides all 15 unordered pairs classified by primary failure mode under the admissibility criterion of Section 4.3 (Collapse Admissibility Criterion).

Reviewer #2 summarized the paper and noted:

1.0) It appeared at first that this paper would address in more detail something I flagged as an interesting direction opened up by the other one -- how to axiomatize certain relationships and entities so as to leave room for subsequent, potentially divergent, specializations. However this is not what is done here.

1.0) RESPONSE: Agreed. In the revision, Section 6 (Interpretation of the Result), specifically Section 6.6 (Implications), now makes explicit how the substrate is axiomatized so as to support divergent specialization. In particular, identity conditions and admissible relations are fixed at the level of identity-and-persistence regimes, while causal and normative commitments are externalized to extensions. This separation ensures that extensions may introduce incompatible causal or normative structure through conservative extension without altering the identity or admissibility conditions of substrate-level entities. The six-regime construction therefore functions as an axiomatization pattern: it fixes the minimal identity-and-persistence structure required for stable reference, while leaving causal and normative structure open to divergent specialization.

2.0) Much like the other paper, there is a lot of text spent on, at least apparently, trivial inferences.

Thus, the requirements of section 3 map almost one to one to the necessary persistence regimes of section 4, leaving little to do for a proof. I will discuss the one exception below, but first, I will remark that the list of requirements of section 3, and the list of persistence regimes, are both very sensible.

This is good, but I doubt it is sufficient for a paper in Applied Ontology because it so happens that the distinctions requested by the paper are also observed by all the foundational ontologies I am aware of.

Thus, there is little in the requirements proposed here that seems to be something the ontology engineering community at large has somehow overlooked so far. The exception to the one to one mapping from requirements to persistence regimes is "3. Acted upon referents". It does not explicitly correspond to any of the requirements,

and it is the one place where it would be better to spell out the necessity proof. I have an intuition as to why this category is there -- because it is intuitive for a "non-agentive enduring" category to be in an ontology (and again, foundational ontologies tend to also make such a distinction). But this is not a persistence regime whose necessity is obvious enough to be swept under a proof sketch rug. Let us accept that time indexed occurrents involve obligation bearing entities who are causally responsible or attributed

some agentive role etc. to drive the occurrent; in short, let us accept that obligation bearing entities can be the agents of occurrents. In plainer English, people can be responsible for their actions. But it also seems reasonable to accept that actions can affect other people directly. Therefore, the class of entities that can be the patients (ie the acted on things) of time indexed occurrents overlaps with the class of obligation bearing entities. Having established that, let us postulate further that all physical entites/entities that can be patients of time indexed occurrents are also obligation bearing.

What is logically inconsistent in this postulation?

It is rather animist, sure, but then why is this viewpoint not admissible?

Even more, this animist viewpoint does not contradict in-scope individual statements of obligation that some other, non-animist interpretation would make.

By in-scope individual statement of obligation, I mean a statement made from some viewpoint about particular individuals, asserting that an individual which that viewpoint recognizes as obligation bearing has (or has not) some obligation. This is because the animist viewpoint asserts that a rock could bear obligations;

it is also happy with the rock's list of obligations being empty. All of this makes it so that the decision as to whether there can be physical entities that are not obligation bearing can be postponed to the viewpoints themselves.

I am not arguing that it should, I am just saying that it seems logically possible to do so.

2.0) RESPONSE: Thank you for highlighting this point. The necessity of the acted-upon referent regime (ENR) is indeed the least immediate part of the lower-bound argument, and the revision strengthens this point in two ways.

First, Section 4.2.6 (Enduring Non-Normative Referents) now makes explicit that ENR is not postulated, but induced by the joint satisfaction of R2-R5. These requirements presuppose stable reference to enduring entities that may participate in occurrences or be subject to regulation or inspection, while not thereby bearing obligations or grounding authority. This yields a persistence-and-admissibility profile in which participation and measurement relations are admissible, but normative attribution is not.

Second, Section 4.4.1 (Category error: Obligation-bearing and Enduring non-normative referent) expands the collapse analysis for OBL/ENR. The issue is not whether it is logically possible to treat all enduring entities as obligation-bearing (e.g., under an animist interpretation), but whether such a collapse is admissible under the neutrality and extension-stability constraints. If ENR is identified with OBL, normative attribution relations become structurally admissible for all enduring referents, embedding normative commitments at the substrate level. If, instead, such relations are restricted to a subset of instances, implementations must introduce discriminators (e.g., flags or predicates), thereby reintroducing the distinction at the extension layer.

Appendix A provides a complete enumeration of all pairwise collapses and classifies OBL/ENR as a category error under the admissibility criterion, showing that this is not an isolated case but part of a systematic failure pattern. The necessity of ENR therefore follows from the admissible-relation criterion and the exclusion of hidden regime reintroduction, not from assumptions about which entities can bear obligations.

3.0) As before, what I see in the paper is either trivial or wrong, and merits a reject. On the chance that there is a severe misunderstanding on my part, I rate this as a "major revisions needed".

3.0) RESPONSE: Thank you for the critical assessment and the chance to clarify the work. The revision addresses the concern that the result may appear trivial by making explicit that the paper does not propose new ontological categories, but derives a structural lower bound on identity-and-persistence regimes under the neutrality and extension-stability constraints. While the resulting distinctions resemble those found in foundational ontologies, the contribution is not the inventory of such distinctions, but the demonstration that such distinctions are unavoidable if one requires extension-invariant identity and the exclusion of hidden discriminators under persistent disagreement. To make this explicit, the revision strengthens both the derivation and the collapse analysis, particularly for the least immediate case (acted-upon referents), as detailed below.

To focus the response, here again are two questions.

3.1) Question 1 Can the author spell out the proof for why acted upon entities are

	necessary as a distinct mode of persistence from obligation bearing entities? 3.1) RESPONSE: Yes; Section 4 now provides an explicit derivation of the acted-upon referent regime (ENR) from the joint satisfaction of R2-R5, and an expanded collapse analysis (Section 4.4.1) showing that identifying ENR with obligation-bearing entities violates the admissibility criterion. Specifically, such a collapse either makes normative attribution relations structurally admissible for all enduring referents, embedding normative commitment at the substrate level, or requires discriminators to restrict those relations, thereby reintroducing a hidden regime. The necessity of ENR therefore follows from the admissible-relation criterion rather than assumptions about which entities can bear obligations. 3.2) Question 2 Can the author describe what their persistence modes bring new to the table compared to such classics of foundational ontologies such as the categories of Endurant/Continuant/Object (with typical subclasses such as PhysicalObject and Agent), Perdurant/Occurrent/Event, Quality, Norm, Situation? 3.2) RESPONSE: Yes; the revision clarifies that the contribution is not the introduction of new ontological categories, but the derivation of a minimal construction of identity-and-persistence regimes under the stated constraints. Foundational ontologies often include distinctions corresponding to enduring entities, occurrences, normative structures, etc., but these are typically introduced on conceptual or modeling grounds. In contrast, this paper shows that, under neutrality and extension-invariant identity, such distinctions are structurally forced: any attempt to collapse them either violates extension-invariant identity, produces category error, or reintroduces hidden discriminators. The result therefore provides a constraint on ontology design: it identifies which distinctions are structurally unavoidable for substrates intended to support accountability under persistent disagreement. ----- We believe the revision addresses the reviewers' concerns and makes the contribution of the paper more precise. Again, many thanks to both reviewers. The paper is significantly improved because of their input.
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Substrate Stability Under Persistent Disagreement: Structural Constraints on Individuation in Neutral Ontological Substrates

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Abstract

Modern data systems increasingly operate under conditions of persistent legal, political, and analytic disagreement. In systems that adopt a neutral substrate, shared interpretation cannot be assumed, and representations must preserve stable reference across incompatible extensions. This paper examines the structural consequences of that architectural choice. Treating interpretive neutrality and extension-stable identity as explicit design constraints, the analysis asks what minimal ontological structure is required for entities and their relations to remain referable and comparable under persistent disagreement. Minimality is defined structurally: reductions are admissible only if they do not reintroduce identity-relevant distinctions through hidden roles, flags, or contextual predicates. A conditional lower bound is established: any ontology intended to function as a neutral substrate under persistent disagreement must realize at least six distinct identity-and-persistence regimes. A construction with exactly six regimes satisfies the stated requirements without embedding causal or normative commitments in

the substrate. Together these results establish a tight structural bound on entity individuation in neutral substrates.

Preview of Main Result. *Under explicit neutrality and stability constraints, attempts to simplify ontological structure by collapsing distinctions reintroduce those distinctions as hidden interpretive mechanisms. Excluding such hidden regimes implies six distinct identity-and-persistence regimes required for accountability under persistent disagreement: fewer do not satisfy the stated requirements, and additional regimes introduce no further structural capacity.*

Informally. Given a neutral substrate capable of supporting persistent interpretive disagreement, entities and relations must remain stably referable across incompatible interpretive frameworks. This requirement constrains how entities may be individuated. Attempts to simplify the individuation structure by collapsing categories reintroduce those distinctions as hidden interpretive mechanisms. Conversely, a construction with six identity-and-persistence regimes provides sufficient structure to preserve stable reference without embedding causal or normative commitments in the substrate. Together these results establish a tight structural bound of six regimes.

Keywords: formal ontology; identity and persistence; neutral substrates; extension stability; ontological neutrality; accountability; causal commitment; normative commitment;

1 Introduction: Structure Under Disagreement

Modern data systems increasingly operate across settings characterized by persistent legal, political, and analytic disagreement, where no single interpretive authority can be assumed. Three recurring settings make the structural requirements concrete. First, in multi-jurisdiction regulatory reporting, a single corporate entity operates under overlapping legal regimes that individuate obligations, violations, and responsible parties differently;

audit trails must remain comparable across jurisdictions without presupposing which regime governs. Second, in legislative history tracking, statutory authority is amended, delegated, and contested over time, requiring stable reference to normative structures independently of the occurrences that enact or challenge them. Third, in cross-institutional provenance analysis, research or administrative records are produced, transferred, and re-analyzed across institutions whose classificatory and causal assumptions diverge, yet must remain referable as the same records across those divergent frameworks. These cases are not offered as empirical evidence; they function as structural diagnostics that make concrete the requirements of Section 3.

Upper ontologies and general-purpose representational frameworks including BFO (Arp et al. 2015a), DOLCE (Gangemi et al. 2002), and PROV-O (Lebo et al. 2013) are expressive and well-motivated for their intended purposes. They support rich domain modeling, causal attribution, and normative classification.

However, the design objective considered in this paper is different. The goal is not representational adequacy within a shared interpretive framework, but substrate stability across incompatible interpretive contexts. This shifts the design problem from domain modeling to the individuation of entities within a neutral substrate. When ontologies are deployed as shared substrates under persistent disagreement without additional constraints, commitments such as what counts as an agent, what individuates an event, and what constitutes an authority, may become sites of interpretive conflict rather than stable reference points.

No single interpretive authority can be presupposed, and semantic convergence cannot be enforced without excluding legitimate participants or perspectives. Nevertheless, shared use of data remains necessary: entities must be referable, actions traceable, and records comparable across frameworks that may diverge in assumptions, methods, or evaluative standards. Recent work is consistent with the view that systems without explicit structural commitments produce unstable and context-dependent categorical behav-

ior (Köhler and Neuhaus 2025), further motivating the need for substrates that enforce stable reference structurally rather than relying on interpretive consistency.

A natural response might be to reduce ontological commitment by collapsing distinctions and representing entities, occurrences, authorities, and records as a single undifferentiated type, differentiated only by role predicates (e.g., *isAgent*, *isOccurrence*, *isAuthority*) or using contextual flags supplied at query or extension time. This achieves surface parsimony but not structural neutrality. Such encodings relocate essential distinctions from the substrate to interpretive mechanisms that are not represented at the substrate level, and whose consistent application requires precisely the shared commitments that persistent disagreement denies. We refer to these as *hidden regimes*: structural discriminations that are operationally necessary but ontologically unacknowledged. Such encodings make referential stability contingent on interpretive consensus that cannot be assumed under persistent disagreement, and are therefore incompatible with the optimization criterion adopted here.

This analysis identifies a design objective that is distinct from the general problem of ontology construction. We adopt a neutrality framework developed in prior work (Anonymous 2026), treating ontological neutrality as a structural design constraint. Neutrality is understood as interpretive non-commitment at the substrate level combined with stability under incompatible extension: a substrate must permit independent extensions to introduce causal or normative structure without requiring revision of the underlying representation. Explanatory and evaluative structure is externalized; the substrate is constrained to remain pre-causal and pre-normative.

The substrate is not intended to adjudicate causal or normative claims. Its role is limited to preserving referential comparability across independently evolving extensions. All causal explanation, normative evaluation, and domain-specific interpretation are delegated to extension layers.

Central to this constraint is the notion of stable reference, which in turn constrains how entities may be individuated within the substrate. Stable

reference in this work concerns regime-level persistence conditions, not immunity to reinterpretation at extension layers. The precise assumptions defining persistent disagreement and extension without coordination are stated in Section 2. Given stable reference as a structural constraint, minimality cannot be defined simply as a reduction in primitive count.

Apparent simplifications may achieve surface parsimony by relocating essential distinctions into roles, flags, contextual typing, or other interpretive mechanisms. Such encodings do not eliminate structure; they merely relocate it to extension layers that presuppose shared interpretive commitments. Under persistent disagreement, those commitments cannot be assumed. Minimality therefore requires that all operationally necessary identity-and-persistence distinctions be explicit and structurally enforced at the substrate level.

Accountability is the specific capacity discussed in this paper. Accountability analysis requires that a neutral substrate maintain stable reference conditions across all entities it represents (R1), and support representation of obligation-bearing entities, normative structures, time-indexed occurrences, applicability contexts, and descriptive records (R2–R6), without embedding causal claims or normative judgments in the substrate itself. These are introduced as representational capacities whose simultaneous support places structural constraints on any neutral substrate. Whether they can be realized within fewer identity-and-persistence regimes is addressed in Section 4; the question is whether a neutral substrate can support them at all, and if so, with what irreducible structural commitments. The diagnostic cases introduced above (multi-jurisdiction regulatory reporting, legislative history tracking, and cross-institutional provenance) recur throughout the paper as structural illustrations.

The central contribution of this paper is a constraint result: a conditional lower bound on ontological structure under the design objectives stated above. Under explicit constraints of neutrality, stability under incompatible extension, and exclusion of hidden structural discriminations, we show that any substrate capable of supporting accountability under persistent disagree-

ment must realize at least six distinct identity-and-persistence regimes. We further show that a construction realizing exactly six such regimes suffices without introducing hidden or excess structure. The result is a constraint on what is structurally possible under a specific optimization objective, not a proposal for a universal ontology. Alternative objectives, such as interpretive convergence, domain-specific efficiency, or context-dependent substrate classification, yield different tradeoffs and are not addressed here.

The remainder of the paper proceeds as follows. Section 2 states the assumptions, neutrality constraints, and optimization criterion. Section 3 formalizes the structural requirements imposed by accountability analysis. Section 4 derives the lower bound. Section 5 presents a construction achieving this bound. Section 6 discusses the scope and implications. Section 7 situates the work relative to existing frameworks. Section 8 concludes.

2 Assumptions and Design Constraints

This section fixes the problem setting for the remainder of the paper. All results that follow are conditional on the assumptions stated here. The purpose is not to argue that these assumptions are universally correct, but to make explicit the constraints under which a neutral substrate for accountability is possible. These constraints determine how entities may be individuated within such a substrate.

2.1 Persistent Disagreement and Extension Without Coordination

We assume environments in which disagreement is not an anomaly to be resolved, but a persistent condition. Legal interpretation, policy evaluation, and analytic methodology may remain incompatible across communities and over time. Accordingly, no single interpretive authority or shared semantic framework is assumed at the substrate level.

We further assume that extensions to the substrate may occur without coordination. Independent actors may add explanatory, normative, or analytic structure, and such extensions must not require revision of the substrate itself. This assumption does not rely on global semantic agreement, centralized governance, or negotiated meaning as preconditions for interoperability. Under these assumptions, substrate-level admissibility must not depend on interpretive agreement among extensions.

2.2 Stability Under Persistent Disagreement

Stable reference in this work concerns regime-level identity and persistence conditions, not immunity to transformation or versioning at interpretive layers. The substrate must preserve the referential integrity of entities and records even as their interpretation evolves in incompatible ways across extensions. Persistent disagreement, as used in this work, is defined by four conditions:

- Extensions may introduce incompatible causal accounts.
- Extensions may introduce incompatible normative attributions.
- Extensions cannot be forced into convergence.
- Substrate revision is not permitted to resolve interpretive conflict.

Under these conditions, the substrate must remain interpretively agnostic while preserving referential stability across extensions. Stability is a property of the identity-and-persistence regime that governs individuation, rather than any particular token or interpretive assignment within it.

2.3 Neutrality Constraint

This work assumes substrate neutrality and stability as a design constraint rather than as a philosophical claim. A neutral substrate does not assert why events occur, whether outcomes are desirable, or how actions ought to

be evaluated. Instead, it provides stable referents and structural relations upon which external interpretive layers may operate.

Neutrality further requires stability under incompatible extension at the level of identity conditions and admissible relations. If two extensions disagree in interpretation, evaluation, or explanation, that disagreement must not force reclassification of entities, alteration of identity conditions, or revision of the substrate. Any representation whose admissibility conditions depend on interpretive agreement does not satisfy this constraint.

2.4 Stable Reference Requirement

For a substrate to support accountability, entities must remain referable over time and across extensions. Identity conditions and admissible relations must not depend on interpretive convergence. If an entity must change its ontological status in order to participate in new relationships or analyses, stable reference is not preserved. This requirement is not satisfied by representations in which identity is determined contextually or retroactively by interpretive layers. The domain consequences of this requirement are illustrated in Section 3 under R1.

This requirement excludes representations in which stability-critical distinctions are encoded only as roles, flags, or contextual predicates. Such mechanisms rely on interpretive conventions supplied by extension layers and therefore permit divergent admissibility judgments under persistent disagreement.

The stability requirement is intentionally strong. Engineering systems may address identity pressure through contextual or negotiated identity schemes, and in such settings strong reference stability may not be attainable in practice. The present analysis instead examines the case where such mechanisms are unavailable and stable reference must be preserved independently of interpretive convergence.

2.5 Optimization Criterion

The optimization objective of this paper is to minimize explicit ontological structure subject to the constraints above. Minimality is not defined as the smallest possible number of primitives in the abstract, but as the smallest number of explicit identity-and-persistence regimes required to preserve neutrality and stable reference.

A proposed reduction constitutes a genuine simplification only if it does not reintroduce the collapsed distinctions through hidden discriminators, interpretive mechanisms, or role predicates.

The minimality criterion adopted here is analytical rather than prescriptive: it is introduced to determine what ontological structure is unavoidable when neutrality and stability under persistent disagreement are treated as design constraints.

Alternative optimization objectives prioritize different tradeoffs, such as interpretive coordination, domain-specific efficiency, or context-dependent classification at the substrate level. The results presented here apply only to substrates optimized for interoperability under persistent disagreement.

These assumptions and constraints determine the design space within which a neutral accountability substrate must operate. The following section states the representational capacities that any substrate must satisfy under these conditions.

3 Structural Requirements for Neutral Substrates

Given the assumptions and optimization criterion fixed in Section 2, the central question is not which ontology is most expressive or convenient, but which structural distinctions are unavoidable. We therefore state the representational requirements that an ontological substrate must satisfy in order to support accountability analysis under persistent disagreement.

These requirements are structural rather than interpretive. They do not prescribe how accountability should be understood, evaluated, or enforced; instead, they specify what must be representable in a stable and neutral manner.

The requirements are framed as necessary capacities. They are not derived from any particular domain, policy area, or theory of governance, but from the minimal conditions under which accountability claims can be stated, inspected, and compared without embedding causal or normative commitments in the substrate.

Before deriving the lower bound, we make explicit what is meant by an *identity-and-persistence regime*.

Definition 3.1. Identity-and-Persistence Regime An identity-and-persistence regime specifies the conditions under which entities remain the same entities over time and across incompatible extensions, together with the relations they may admit without altering their identity conditions. Identity-and-persistence regimes therefore represent distinct modes of individuation within a neutral substrate.

Two entities belong to the same regime if and only if they share the same stability-critical identity criteria, persistence conditions, and admissible relations under the neutrality and stability constraints adopted here.

Regimes are not introduced as ontological kinds or domain categories. They are structural distinctions forced by the requirement that identity and admissibility remain invariant under persistent disagreement.

The requirements enumerated below differ in scope. R1 is a cross-cutting constraint on individuation: it applies to every entity representable in the substrate, regardless of which other requirements govern its regime. R2–R6 each specify representational capacities not entailed by R1 alone.

3.1 R1: Stable Identity and Persistence (Global Constraint)

Requirement. The substrate must support stable reference to all entities it represents over time.

R1 operationalizes the Stable Reference Requirement (Section 2.4) as a representational capacity governing how entities are individuated and tracked within the substrate. Changes in interpretation, role, status, or evaluation must not require reclassification of entities or modification of their identity criteria. Representations that determine identity contextually or retroactively through interpretive layers fail to satisfy R1: their identity conditions are not invariant under extension and therefore do not constitute stable reference for accountability purposes. In addition, admissible relations must be invariant under extension: whether a relation is well-formed between entities of given regimes must not depend on interpretive framework or extension-layer agreement. No relation may determine or alter the identity conditions of the entities it connects, nor introduce or encode identity criteria through participation in relations.

In cross-institutional provenance, a research record transferred between institutions must remain referable as the same record regardless of how each institution classifies, annotates, or re-analyzes it. If identity conditions shift when an institution introduces a new classificatory scheme, the provenance chain breaks and accountability for the record’s content and custody cannot be maintained. R1 is therefore non-optional wherever records must persist as stable referents across frameworks that may individuate them differently.

3.2 R2: Obligation-Bearing Capacity

Requirement. The substrate must support representation of entities that can bear obligations or responsibilities independently of particular occurrences.

Such entities must be referable independently of the particular occurrences

in which obligations are fulfilled, violated, or discharged. This requirement does not presuppose any substantive theory of normativity. It requires only that obligation-bearing entities be representable as enduring referents whose identity does not depend on specific events.

In multi-jurisdiction regulatory reporting, a corporate entity must be referable as an obligation-bearer across legal regimes that may differ in how they define corporate personhood, subsidiary responsibility, or delegated liability. If the substrate cannot represent obligation-bearing entities independently of the specific events in which obligations are fulfilled or violated, cross-jurisdiction audit comparisons require interpretive agreement on the nature of the bearer itself, precisely the agreement that cannot be assumed under persistent disagreement. R2 is therefore non-optional wherever obligation-bearers must be referable across legal or institutional frameworks that individuate corporate personhood, subsidiary responsibility, or delegated liability differently.

3.3 R3: Normative Reference Without Execution

Requirement. The substrate must distinguish between the existence of authority or obligation and the occurrences that execute, comply with, or violate it.

Normative or regulatory structures must therefore be representable independently of the occurrences that enact, comply with, or violate them. This distinction is necessary to support accountability over time, including cases of delayed enforcement, partial compliance, amendment, or contested application. Encoding normative reference solely through occurrences collapses authority into execution and undermines stable reference to governing structures.

In legislative history tracking, a statutory authority must be referable as a normative structure independently of the legislative acts that enacted it, the administrative actions that applied it, and the judicial decisions that interpreted it. If normative reference is encoded solely through occurrences,

a statute that has been amended but not yet enforced, or contested but not yet resolved, has no stable substrate-level referent. R3 is therefore non-optional wherever accountability requires distinguishing compliance with authority-as-written from compliance with authority-as-executed, including cases of amendment, delayed enforcement, or contested application.

3.4 R4: Time-Indexed Occurrence

Requirement. The substrate must support representation of occurrences individuated by temporal realization and provenance.

Occurrences must be individuated by temporal realization and provenance, and must remain distinct from enduring entities. This requirement applies to all discrete occurrences relevant to accountability, including actions, filings, inspections, or transactions. The list is illustrative; the requirement extends to any occurrence for which temporal individuation and provenance are necessary for stable reference. Occurrences are not required to carry interpretive or evaluative meaning; they must assert only that something occurred at a particular time and place.

In multi-jurisdiction regulatory reporting, a filing, inspection, or transaction must be individuated by when and where it occurred, independently of how different jurisdictions classify or evaluate it. Two jurisdictions may disagree on whether a given action constitutes compliance, but both must be able to refer to the same occurrence as their object of evaluation. Without substrate-level individuation of time-indexed occurrences, that shared reference is unavailable and cross-jurisdiction comparison reduces to a dispute about what happened rather than about how to evaluate it. R4 is therefore non-optional wherever cross-jurisdiction accountability requires shared reference to occurrences independently of how those occurrences are classified or evaluated.

Requirements R2, R3, and R4 are mutually irreducible as representational capacities. R2 requires stable reference to an enduring bearer; R3 requires

stable reference to a governing normative structure; R4 requires stable reference to a discrete, time-indexed occurrence. Satisfying any one does not entail satisfying the others. None of the three capacities is definable in terms of the remaining two without either circularity or the collapse of distinctions that R1 requires to remain stable across incompatible extensions.

3.5 R5: Applicability and Scope

Requirement. The substrate must support representation of applicability contexts that scope where, to whom, or under what conditions a normative structure applies.

It must be possible to state where, to whom, or under what conditions a normative or regulatory structure applies, without collapsing such contexts into either obligation-bearing entities or physical loci. Applicability contexts may be nested, overlapping, or partially coincident. They must therefore be representable as stable referents in their own right, rather than as incidental attributes whose interpretation may vary across extensions.

Applicability contexts cannot be reduced to properties of the R3 normative structures they scope: a single normative structure may admit different applicability contexts under different jurisdictional frameworks, and collapsing scope into the structure would require its identity conditions to vary with those frameworks, violating R1. Nor can applicability contexts be reduced to obligation-bearing entities (R2), since scope specifies the domain over which authority applies rather than identifying any particular bearer.

In multi-jurisdiction regulatory reporting, normative structures apply to different sets of entities depending on jurisdiction, and those jurisdictions may be nested, overlapping, or contested. If applicability contexts cannot be represented as stable referents in their own right, the scope of an obligation can only be recovered by interpreting the obligation itself which reintroduces the shared interpretive assumptions the substrate is designed to avoid. R5 is non-optional wherever accountability claims must be scoped without

presupposing agreement on which normative framework or authority governs.

R2–R5 together establish the referential structure required for accountability analysis: obligation-bearing entities (R2), governing normative structures (R3), time-indexed occurrences (R4), and the applicability contexts that scope those structures (R5). These capacities make it possible to represent who is responsible, which authority governs, what occurred, and where that authority applies. They do not, however, establish stable reference to the descriptive records or indicators associated with those entities and occurrences.

3.6 R6: Descriptive Indicators Without Causal Commitment

Requirement. The substrate must support representation of descriptive indicators or records that characterize states, performance, or outcomes associated with entities or occurrences without asserting causal or normative claims.

R6 is not entailed by R1–R5. Stable reference to obligation-bearing entities (R2), normative structures (R3), time-indexed occurrences (R4), and applicability contexts (R5) does not establish stable reference to descriptive records associated with those referents. Descriptive indicators have individuation conditions that are independent of temporal realization (R4), do not assert or constitute normative authority (R3), do not identify obligation-bearers (R2), and do not scope the application of normative structures (R5). They must therefore be specified by a distinct requirement.

Such indicators must be representable without asserting causal relations. This requirement does not preclude the representation of causal claims at extension layers. This requirement is necessary to support longitudinal comparison and outcome inspection while preserving neutrality. Collapsing indicators into occurrences or into interpretive annotations eliminates stable reference to descriptive records over time and across analytic frameworks.

In cross-institutional provenance, performance indicators and outcome records must remain comparable across institutions whose causal and normative frameworks differ, for instance, where one institution models outcomes in terms of intervention efficacy and another in terms of systemic risk factors. If descriptive indicators are encoded as occurrences, they carry implicit individuation conditions tied to causal assumptions; if encoded as interpretive annotations, they are unstable under extension. R6 is non-optional wherever longitudinal comparison of records must be preserved without committing the substrate to any particular explanatory or evaluative framework.

3.7 Summary

Together, R1–R6 specify the minimal representational capacities required for an accountability substrate operating under persistent disagreement. R1 is a cross-cutting stability constraint that applies to all entities the substrate must represent. R2–R5 specify four distinct representational capacities that are mutually irreducible: no one of them is entailed by or definable in terms of the others. R6 specifies a further capacity not entailed by R1–R5. These requirements do not impose a particular ontological partition; they define the structural constraints under which any such partition must operate. The following section uses R1–R6 to derive a lower bound on the number of distinct identity-and-persistence regimes a substrate must realize in order to satisfy them simultaneously.

4 Necessity: A Lower Bound on Ontological Structure

The previous section specified the structural requirements for substrates under neutrality and extension-stable identity. We now use those requirements to derive a lower bound on the individuation structure that any such substrate must realize.

The result is a necessity claim: if a substrate is to support accountability under persistent disagreement while maintaining neutrality and stable reference, some minimum number of distinct identity-and-persistence regimes is unavoidable. The lower bound is established by showing that certain representational capacities cannot be realized within a single regime without violating extension-invariant identity or reintroducing hidden discriminators.

The regimes introduced below are persistence-and-admissibility profiles induced when the requirements of Section 3 must be satisfied simultaneously under the neutrality and stability constraints fixed in Section 2.

4.1 Goal and Proof Strategy

The goal of this section is to prove a numeric lower bound on the number of identity-and-persistence regimes required to satisfy R1–R6 simultaneously under the optimization criterion fixed in Section 2.

The argument has three steps.

First, we extract from R2–R6 (subject to R1) a set of structurally distinct persistence-and-admissibility profiles that must be realizable in any compliant substrate. These profiles correspond to candidate identity-and-persistence regimes forced by the requirements.

Second, we state a collapse admissibility criterion: a proposed identification of two regimes is admissible only if it preserves extension-invariant identity and admissible relations without introducing interpretive flags, roles, or context-dependent predicates at the extension layer.

Third, we show that any collapse among the forced profiles violates the admissibility criterion through one of three failure modes: identity instability, category error, or hidden regime reintroduction. It follows that the substrate must realize at least as many distinct regimes as there are forced profiles, establishing the lower bound.

4.2 Induced Identity-and-Persistence Regimes

We now determine which identity-and-persistence regimes are forced by R2–R6 when satisfied simultaneously under R1’s requirement of extension-invariant identity and admissible relations.

An identity-and-persistence regime is characterized by (i) persistence conditions and (ii) the class of relations admissible without reclassification.

When the requirements are satisfied simultaneously under R1, their persistence and admissibility constraints give rise to distinct candidate identity-and-persistence profiles. We extract these profiles in turn.

4.2.1 Profile induced by R2: Obligation-bearing persistence

R2 requires representation of entities that bear obligations independently of particular occurrences. Such entities must persist across compliance, violation, delegation, and discharge. Their admissible relations include normative attribution without altering identity. This induces an initial distinct persistence-and-admissibility profile: obligation-bearing.

4.2.2 Profile induced by R3: Normative-structure persistence

R3 requires representation of normative or regulatory structures independently of their execution. Such entities persist across enactment, amendment, interpretation, and repeal. Their admissible relations include structural modification without collapsing into occurrences. This induces a second distinct profile: normative-structure.

4.2.3 Profile induced by R4: Time-indexed occurrence persistence

R4 requires representation of occurrences individuated by temporal realization and provenance. Such entities do not persist through modification

but are fixed by their time-indexed realization. This induces a third distinct persistence profile incompatible with enduring entities: time-indexed occurrence.

4.2.4 Profile induced by R5: Applicability-context persistence

R5 requires representation of scopes or applicability contexts as enduring entities. Such entities admit nesting, overlap, and conditional scoping without collapsing into either authority or bearer entities. This induces a fourth structurally distinct profile: applicability-context.

4.2.5 Profile induced by R6: Descriptive-record persistence

R6 requires representation of descriptive indicators without causal or normative commitment. Such entities admit measurement and comparison relations without implying execution or authority. This induces a fifth distinct profile: descriptive-record.

4.2.6 Enduring Non-Normative Referents

Satisfaction of R2–R5 presupposes stable reference to enduring entities that are neither obligation-bearing nor authority-bearing, yet may be regulated, inspected, or otherwise affected by occurrences.

Such entities persist independently of normative attribution and independently of time-indexed realization.

Their persistence conditions are those of enduring referents whose identity is not altered by normative attribution, structural amendment, or temporal realization.

Their admissible relations include participation in occurrences as patients or objects of action, as well as measurement and description relations, without

implying normative authority or responsibility.

Profiles are individuated by the conjunction of persistence conditions and admissible-relation classes. Two candidate regimes are distinct whenever either their persistence criteria or their admissible-relation profiles differ.

For example, although ENR and REC both admit measurement and description relations, their persistence conditions and participation roles differ, yielding distinct identity-and-admissibility profiles.

This profile is not independently required by any single requirement. However, satisfying R2–R5 simultaneously presupposes stable reference to enduring entities that may participate in occurrences or be subject to regulation or inspection without thereby bearing obligations or grounding authority.

Such entities must admit participation and measurement relations without admitting normative attribution relations.

This admissibility profile is incompatible with that of obligation-bearing entities, whose structural relations include normative attribution. If they were identified with obligation-bearing entities, normative attribution relations would become structurally admissible for all enduring referents, or implementations would require a discriminator to block such attributions. Either outcome violates the admissibility criterion introduced below.

Accordingly, the representational capacities required by R2–R5 induce a distinct persistence-and-admissibility profile for enduring non-normative referents.

This yields a sixth distinct persistence-and-admissibility profile: enduring non-normative referents.

4.2.7 Resulting Candidate Partition

The induced profiles are pairwise non-equivalent under R1’s stability constraint. They differ either in persistence conditions or in admissible-relation

classes.

Accordingly, any substrate satisfying R1–R6 must realize at least six distinct identity-and-persistence regimes. For reference, we fix the following abbreviations for these regimes:

1. **OBL: Obligation-bearing entity:** an enduring referent that may bear obligations or responsibilities and remain identifiable as a party across time.
2. **NOR: Normative structure:** an enduring referent that grounds obligations or permissions without being tied to any single occurrence.
3. **OCC: Time-indexed occurrence:** an entity individuated by its temporal realization and provenance.
4. **CTX: Applicability context:** a referent that scopes where, to whom, or under what conditions authority applies, and that may be nested or overlapping.
5. **REC: Descriptive record:** a referent that reports measured or derived properties without asserting causal or normative conclusions.
6. **ENR: Enduring non-normative referent:** an enduring entity that may be regulated, inspected, or otherwise affected by occurrences, but does not itself ground obligations or bear normative responsibility.

4.3 Collapse Admissibility Criterion

To establish a lower bound, we consider whether any two of the induced identity-and-persistence regimes can be collapsed into a single regime without violating the constraints of Section 2.

A collapse of two regimes is admissible only if the merged representation realizes a single extension-stable identity and persistence profile and supports a unified class of admissible relations without introducing interpretive flags,

role predicates, or contextual discriminators at the extension layer, which would reintroduce regime distinctions contrary to the neutrality and extension-stability constraints of Section 2.

Under R1, identity and admissibility must be invariant across incompatible extensions. Accordingly, a collapse is inadmissible if it requires any of the following:

- **Identity instability:** the merged regime requires incompatible persistence criteria, so that identity judgments vary across extensions.
- **Category error:** the merged regime structurally admits relations inappropriate to some of its instances, requiring interpretive blocking rules.
- **Hidden regime reintroduction:** the original distinction is recovered through flags, roles, or contextual predicates, thereby reintroducing the regime at the extension layer.

Any collapse exhibiting one of these failure modes violates either extension-invariant identity or the neutrality criterion excluding hidden discriminators.

4.4 Pairwise Collapse Analysis

There are finitely many unordered pairs among the induced regimes. We consider each class of collapse and show that every such identification violates the admissibility criterion by instantiating one of the three failure modes defined above.

4.4.1 Category error: Obligation-bearing and Enduring non-normative referent

Suppose an obligation-bearing entity and an enduring non-normative referent are collapsed into a single regime.

The merged regime must admit a unified class of admissible relations.

Obligation-bearing entities admit normative attribution relations such as *hasObligation* or *isResponsibleFor* without reclassification.

If those relations are structurally admissible for all instances of the merged regime, then every enduring referent becomes eligible for normative attribution in principle, embedding a normative commitment at the substrate level.

If, instead, normative attribution is restricted to only some instances, implementations must introduce a discriminator such as *isObligationBearer* to block inappropriate attributions. The original distinction is thereby reintroduced at the extension layer.

In either case, the collapse is inadmissible.

4.4.2 Identity instability: Obligation-bearing and Normative structure

Suppose an obligation-bearing entity and a normative structure are collapsed into a single regime.

Obligation-bearing entities persist across compliance, violation, and delegation. Normative structures persist across enactment, amendment, and repeal.

These persistence criteria are non-equivalent. No single identity condition can preserve both without making persistence judgments vary across extensions.

The collapse therefore violates extension-invariant identity under R1.

4.4.3 Category error: Normative structure and Applicability context

Suppose a normative structure and an applicability context are collapsed into a single regime.

Normative structures admit structural modification relations such as enactment, amendment, delegation, and repeal. Applicability contexts instead admit scoping relations such as nesting, overlap, and conditional refinement.

If the two are identified, the merged regime must admit both modification-type and scoping-type relations as structurally admissible. However, these relation classes serve fundamentally different roles. Amendment is a structural modification that preserves the identity of a normative structure across change. Whereas nesting and overlap are relations among scopes that organize applicability conditions without modifying normative content.

No single admissible-relation profile can unify these relation classes. Either the relations are admitted uniformly, producing category error for some instances, or their admissibility must be restricted contextually, reintroducing the distinction through a hidden discriminator.

If all relations are admitted uniformly, entities that are merely applicability scopes become amendable normative structures, producing category error.

If relation admissibility is restricted by a predicate such as *isNormative* or *isScope*, the original distinction is reintroduced as a hidden regime at the extension layer.

The collapse is therefore inadmissible.

4.4.4 Hidden regime reintroduction: Normative structure and Descriptive record

Suppose a normative structure and a descriptive record are collapsed into a single regime.

Normative structures ground obligations and admit normative evaluation, enforcement, and modification. Descriptive records report measured or derived properties without asserting causal or normative commitments.

If both are identified, the merged regime must admit a unified admissible-

relation profile. If normative evaluation relations are structurally admissible for all instances of the merged regime, then purely descriptive records become eligible for normative grounding, embedding normative commitments at the substrate level.

If, instead, normative evaluation is restricted to a subset of instances, implementations must introduce a discriminator such as *isNormative* or *isDescriptive* to distinguish grounding entities from descriptive ones. Such a discriminator reintroduces the original distinction at the extension layer, violating the admissibility criterion and the neutrality constraint.

The collapse is therefore inadmissible.

4.4.5 Summary of representative collapse failures

Collapsing time-indexed occurrences with enduring non-normative referents produces identity instability. Occurrence identity is fixed by temporal realization, whereas enduring referents persist independently of time-indexed realization. Accordingly, no unified persistence criterion can satisfy both.

Table 1 summarizes representative pairwise collapses and the failure mode each induces under the admissibility criterion defined in Section 4.3.

These three failure modes are exhaustive relative to the admissibility criterion. Since the admissibility criterion requires exactly three conditions to hold jointly (extension-invariant identity, uniform admissibility of relations for the merged regime, and neutrality with respect to hidden discriminators) any collapse can violate admissibility only by failing one of these conditions. Identity instability corresponds to violation of the first, category error to violation of the second, and hidden regime reintroduction to violation of the third.

The remaining pairwise collapses instantiate the same three failure modes: identity instability, category error, or hidden regime reintroduction. Under the admissibility criterion of Section 4.3, no collapse eliminates a regime

Table 1: Representative collapse failures under neutrality and stability constraints

Collapsed regimes	Failure mode	Reason for inadmissibility
OBL / ENR	Category error	Normative attribution relations appropriate to obligation-bearing entities would become structurally admissible for all enduring referents, misclassifying non-normative entities as potential obligation-bearers.
OBL / NOR	Identity instability	Persistence as a responsible party and persistence as a normative structure require incompatible identity criteria across amendment and restructuring.
NOR / CTX	Category error	Modification relations (enactment, amendment, repeal) conflict structurally with scoping relations (nesting, overlap, conditional refinement); no unified admissible-relation profile can serve both without misclassifying instances of one type as the other.
NOR / REC	Hidden regime reintroduction	Normative grounding and purely descriptive reporting admit incompatible admissibility profiles; implementations reintroduce <i>isNormative</i> or <i>isDescriptive</i> flags to recover the distinction.
OCC / ENR	Identity instability	Time-indexed individuation criteria conflict with extension-invariant persistence required for enduring entities.

without violating neutrality or stable reference (see Appendix 8 for the complete enumeration of all fifteen pairs).

The following section establishes that this bound is tight: a construction realizing exactly six regimes satisfies R1–R6 without hidden discriminators or additional identity categories, confirming that six is the minimum achievable structure under the stated constraints.

4.5 Lower-Bound Result

Lemma 4.1 (Lower bound on identity-and-persistence regimes). *Under the assumptions of Section 2, any ontology satisfying requirements R1–R6 and the collapse admissibility criterion of Section 4.3 must realize at least six distinct identity-and-persistence regimes.*

Proof sketch. Section 4.2 shows that satisfying R1–R6 under extension-invariant identity induces six pairwise non-equivalent persistence-and-admissibility profiles, each corresponding to a candidate identity-and-persistence regime.

Any ontology realizing fewer than six regimes must identify at least two of these profiles within a single regime. By the collapse admissibility criterion (Section 4.3), every such identification produces either identity instability, category error, or hidden regime reintroduction, as demonstrated by exhaustive pairwise analysis (Section 4.4 and Appendix 8).

Each of these outcomes violates the neutrality or stability constraints fixed in Section 2.

Therefore no ontology with fewer than six regimes can satisfy R1–R6 simultaneously. $\square$

4.6 Scope of the Necessity Result

The necessity result is conditional on the assumptions of Section 2. It does not claim that fewer regimes are impossible in general, but that they are incompatible with the optimization objective fixed in Section 2. Ontologies optimized for local agreement, negotiated semantics, or expressive flexibility may legitimately employ fewer explicit distinctions by relying on hidden or interpretive mechanisms.

Within the stated constraints, the result establishes a lower bound: any neutral substrate capable of supporting accountability under persistent disagreement must realize at least six distinct identity-and-persistence regimes. The following section shows that this bound is tight by exhibiting a construction that satisfies all requirements with a minimal set of six regimes.

5 Sufficiency: A Six-Regime Construction

5.1 The Six Identity-and-Persistence Regimes

This section exhibits a construction realizing the six identity-and-persistence regimes derived in Section 4. No additional regimes are introduced, and none of the induced regimes are merged or weakened.

Each regime is implemented with the identity conditions and admissible-relation profile specified in the derivation. The construction fixes these regimes as pairwise disjoint, with extension-invariant persistence criteria, and without interpretive flags or role-based discriminators.

We now specify a minimal set of typed relations that connects these regimes while preserving neutrality and stable reference.

5.2 Admissible Relations

The construction introduces the following typed relations. Together these relations realize the admissibility profiles required to satisfy R2–R6. Each requirement is realized by at least one listed relation. The coverage mapping is made explicit in Section 5.3.

- authority grounding and delegation:

$$\textit{enacts}, \textit{issues} : \text{OBL} \rightarrow \text{NOR}$$

- participation and party status:

$$\textit{party-to} : \text{OBL} \rightarrow \text{NOR}$$

$$\textit{involves} : \text{OCC} \rightarrow \text{OBL}$$

$$\textit{has-patient} : \text{OCC} \rightarrow \text{ENR}$$

- execution under authority:

$$\textit{occurs-under} : \text{OCC} \rightarrow \text{NOR}$$

- applicability:

$$\textit{applies-in} : \text{NOR} \rightarrow \text{CTX}$$

$$\textit{refines} : \text{CTX} \rightarrow \text{CTX}$$

$$\textit{overlaps} : \text{CTX} \leftrightarrow \text{CTX}$$

- description:

$$\textit{measures} : \text{REC} \rightarrow (\text{OBL} \mid \text{ENR} \mid \text{CTX})$$

$$\textit{observes} : \text{REC} \rightarrow \text{OCC}$$

Relations in this construction are structural and descriptive. They assert reference, participation, scoping, or provenance, but do not encode causal mechanism or normative validity.

Authority grounding and delegation records that an obligation-bearing entity enacts or issues a normative structure, without asserting validity or enforceability.

Participation relations record that an occurrence is associated with obligation-bearing entities via *involves* and with enduring non-normative referents via *has-patient*.

The *measures* relation associates a descriptive record with an enduring referent, while *observes* anchors a descriptive record to a time-indexed occurrence. Neither relation asserts accuracy, causal relevance, or normative significance.

Occurrences serve as temporal and provenance anchors for participation and descriptive relations. Descriptive records target enduring referents whose identity persists across incompatible extensions.

This relation set suffices to satisfy R1–R6 without introducing additional identity regimes, role-based typing, or extension-layer discriminators.

Temporal and provenance information attach only to OCC and REC via typed attributes or admissible relations. These annotations do not introduce new identity criteria or additional identity-bearing categories.

5.3 Satisfaction of Structural Requirements

The relation set defined above satisfies R1–R6 by the following coverage mapping:

- **R1 (Stable identity):** Each regime fixes an extension-invariant identity-and-persistence criterion. No relation in the construction propagates persistence criteria across regime boundaries.
- **R2 (Obligation-bearing capacity):** *enacts*, *issues*, *party-to*, and *involves* realize normative attribution for OBL without reclassification.
- **R3 (Normative reference without execution):** *enacts*, *issues*, and *applies-in* ground NOR independently of any OCC instance.
- **R4 (Time-indexed occurrence):** *involves*, *has-patient*, *occurs-under*, and *observes* anchor OCC to its participants and governing normative structure without altering their identity.

- **R5 (Applicability and scope):** *applies-in* scopes NOR within CTX; *refines* realizes conditional nesting; *overlaps* realizes non-hierarchical partial scope without requiring containment.
- **R6 (Descriptive indicators):** *measures* associates REC with enduring referents (OBL, ENR, CTX) without asserting causal or normative conclusions.

No requirement introduces a relation that propagates persistence criteria or normative commitments outside the regime for which it is specified. No additional identity regimes or extension-layer discriminators are required.

5.4 Sufficiency Proposition

Proposition 5.1 (Sufficiency of six regimes). *The six-regime construction specified in this section satisfies R1–R6 under the assumptions of Section 2.*

Proof. By construction, each of R2–R6 is realized by at least one typed relation in the construction, and all regimes satisfy R1. The regimes are pairwise disjoint, and the typed relation set connects them only through descriptive, participation, and scoping relations whose admissibility conditions do not propagate persistence criteria across regime boundaries. and the construction introduces no interpretive flags, role-based discriminators, or additional identity regimes. Hence the construction satisfies R1–R6. $\square$

5.5 Tightness of the Bound

Lemma 4.1 establishes that any ontology satisfying R1–R6 must realize at least six distinct identity-and-persistence regimes. Proposition 5.1 shows that the construction presented in this section satisfies those requirements with six regimes.

The six regimes of the construction are pairwise non-equivalent under R1’s stability constraint: each pair differs in persistence conditions, admissible-

relation classes, or both, as established in Section 4.2. Any identification of two regimes would therefore instantiate one of the failure modes defined in Section 4.3, violating either extension-invariant identity or the neutrality constraint.

Thus the lower bound established in Lemma 4.1 is achieved exactly by the present construction. Six regimes are both necessary and sufficient, and the bound is therefore tight.

No compliant construction can realize fewer regimes without violating either extension-invariant identity or the neutrality constraint.

6 Interpretation of the Result

This section clarifies how the lower- and upper-bound results should be understood, and how they should not be understood. The aim is not to defend the result against alternative modeling goals, but to situate it relative to the explicit optimization criterion fixed earlier in the paper.

6.1 Combined Characterization and Scope

Taken together, the necessity and sufficiency results establish a conditional optimality claim. Under the assumptions of Section 2, any ontology intended to function as a neutral substrate for accountability under persistent disagreement must realize at least six distinct identity-and-persistence regimes. Conversely, a construction with a minimal set of six such regimes suffices to meet all stated representational requirements without introducing hidden structure or interpretive commitments.

The result is therefore tight. It does not merely show adequacy or inadequacy under particular conditions; it establishes that six regimes are both necessary and sufficient under the stated optimization objective.

Theorem 6.1 (Tightness of the six-regime bound). *Under the assumptions of Sections 2–3, exactly six identity-and-persistence regimes are necessary and sufficient for a neutral accountability substrate under persistent disagreement.*

Proof. The theorem follows directly from Lemma 4.1 and Proposition 5.1. $\square$

6.2 Minimality Relative to Stability

The result should not be read as a claim about simplicity in an intuitive or engineering sense. Ontologies with fewer explicit entity kinds may be easier to describe or extend within a coordinated interpretive community. However, when such reductions are achieved by encoding stability-critical distinctions as roles, flags, or contextual predicates, total structural complexity is not reduced. Instead, it is displaced into extension layers whose interpretation cannot be held fixed under disagreement.

In this paper, minimality is defined relative to stability under persistent disagreement. *A representation is minimal only if removing explicit structure does not require reintroducing the same distinctions implicitly.* From this perspective, an ontology with fewer explicit kinds but greater interpretive burden is not simpler, but less explicit.

6.3 Why Hidden Regimes Are Excluded

Hidden regimes are often effective where interpretation is shared, enforced, or low-cost to renegotiate. They support flexibility and expressive economy within a single governance framework.

The present analysis excludes such regimes not because they are incorrect, but because they are incompatible with the stated objective. When disagreement is persistent and uncoordinated, distinctions that are not fixed at the substrate level cannot be relied upon to preserve identity or admissible relations. Stability then depends on interpretive convergence, which the

substrate is explicitly designed not to assume.

6.4 Relation to Other Ontological Strategies

Many upper ontologies and modeling frameworks adopt different optimization objectives, including metaphysical coverage, expressive richness, or alignment with natural language or scientific theory. Such frameworks may introduce additional kinds or tolerate contextual typing precisely in order to support richer modeling.

The result presented here neither subsumes nor contradicts those frameworks. It identifies a specific point in the design space: the minimal structure required for a neutral, extension-stable substrate under persistent disagreement. Frameworks optimized for other goals may legitimately occupy different points in that space.

The following comparison is illustrative rather than evaluative. Table 2 shows common representation strategies for these regimes in BFO 2.0 and DOLCE and highlights where regime-level distinctions required for neutrality are not structurally enforced.

The comparison is structural rather than taxonomic; it identifies where regime-level distinctions required for neutrality are left to implementation rather than enforced at the type level.

The table identifies the structural under-constraint that would arise in each case. These under-constraints are not problems in their intended use. They are points where neutral substrate implementations built on those frameworks might need to introduce additional modeling commitments that the present substrate treats as regime-level structure.

The six regimes are not derived from gaps in existing frameworks. Rather, independently motivated requirements reveal where those frameworks leave regime-level distinctions to the implementer.

Table 2: Illustrative representation strategies for the six regimes in BFO 2.0 and DOLCE, with structural under-constraints under neutrality and stability requirements.

Regime	BFO 2.0	DOLCE
ENR (enduring non-normative referent)	Independent Continuant (material object). Clean for material entities; institutional and fiat entities require additional modeling choices not enforced by the framework.	Physical or Social Object (endurant). Social object is first-class, providing a cleaner fit for institutional referents than BFO.
OBL (obligation-bearing entity)	Specifically Dependent Continuant (Role or Disposition) inhering in an Independent Continuant. Obligation-bearing is not a first-class category; it is realized via a role-bearer pair, splitting what the present substrate treats as a single regime.	Quality or property of a Social Object. Obligation-bearing capacity is not a structural first-class distinction; the OBL/NOR boundary remains implementation-dependent.
NOR (non-normative structure)	Generically Dependent Continuant (information artifact, normative document). The substrate-level separation between authority-as-referent (NOR) and authority-as-executed (OCC) is a modeling choice, not structurally enforced.	Social Object (normative structure). Better motivated than BFO, but authority and execution can both be represented as social objects or perdurants without a regime-level firewall between NOR and OCC.
OCC (time-indexed occurrence)	Process or Process Boundary (occurrent). Closest mapping in both frameworks; BFO occurrents are well-individuated temporally.	Perdurant (Event or Process). DOLCE distinguishes achievements from activities, which is richer than required but not incompatible.
CTX (applicability context)	Site (immaterial Independent Continuant) for spatial scope only. Non-spatial applicability contexts (normative jurisdiction, conditional scope) have no stable BFO home and are represented as GDC, conflating CTX with NOR.	Abstract Region or Feature. DOLCE regions are spatiotemporal; normative applicability collapses into Social Object (conflating CTX with NOR) or Quality (conflating CTX with REC).
REC (descriptive indicators without causal commitment)	Generically Dependent Continuant (data item, measurement information entity). Both normative structures (NOR) and descriptive indicators (REC) map to GDC; the stability-critical distinction between them is not explicitly represented at the BFO type level.	Quality (with associated quale or region). DOLCE’s quality theory is the strongest match; the without-causal-commitment constraint is a modeling discipline, however, not structurally enforced.

Table 3 summarizes the systematic under-constraints that arise in both frameworks relative to the neutrality and stability requirements.

Table 3: Systematic under-constraints in BFO 2.0 and DOLCE relative to neutrality and stability requirements.

Under-constraint	BFO 2.0	DOLCE
NOR/REC conflated	Both normative structures and descriptive indicators map to GDC; the stability-critical distinction is not explicitly represented at the type level and left to the implementer.	Both map to Social Object or Quality depending on implementation; the distinction is a modeling discipline, not structurally enforced.
CTX has no structural home	Non-spatial applicability contexts are represented as GDC, effectively conflating CTX with NOR.	Normative applicability collapses into Social Object (conflating CTX with NOR) or Quality (conflating CTX with REC).
OBL/NOR boundary implementation-dependent	The separation between obligation-bearer and normative structure is mediated by the inheres-in relation, not enforced by regime membership; unstable under delegated authority.	The same boundary is mediated by participation relations; regime-level enforcement is absent.

Implementations that resolve these under-constraints silently introduce hidden regimes in the sense defined in Section 4.

6.5 Conditional Scope of the Claim

The lower-bound result is explicitly conditional. If any assumption of Section 2 is rejected, such as neutrality, stability under incompatible extension, or the exclusion of hidden regimes, the optimization criterion is invalidated and the bound no longer applies. We do not claim that these assumptions are universally correct; we claim only that the result follows when they are adopted.

If stability is weakened to allow context-dependent regime membership (e.g.,

role- or viewpoint-relative identity/admissibility), some regimes may be representable as role- or parameter-dependent refinements of fewer base regimes, but doing so shifts structural burden into coordination, versioning, or exchange mechanisms whose admissibility conditions are not fixed at the substrate level and are outside the scope of this analysis; the lower bound therefore does not apply under such weakenings.

Within the stated scope, however, the result is unavoidable. Under the stated assumptions, any attempt to reduce the number of regimes would require sacrificing stability, reintroducing hidden distinctions, or weakening the neutrality constraint. The six-regime structure thus emerges not as a design preference, but as a structural consequence of the problem as posed.

The present analysis does not model state transitions, versioning mechanisms, or exchange protocols. Such structures may be required for practical interoperability, but they operate downstream of the regime-level identity constraints established here.

6.6 Implications

The primary implication of this result is diagnostic rather than prescriptive. It provides a principled way to evaluate claims that an ontology supports neutral interoperability under disagreement. If such a framework relies on implicit roles or context-dependent typing for stability-critical distinctions, the present result explains why those claims cannot be sustained simultaneously under the stated constraints.

Conversely, for designers who share the stated optimization objective, the result identifies which distinctions must be made explicit, and which may safely be deferred to extension layers, in order to preserve neutrality and stable reference across incompatible interpretations. The six-regime structure thus serves both as a lower bound and as a design template for accountability substrates in contested domains.

The resulting regime structure can be understood as an axiomatization pattern for substrates intended to support divergent specialization: identity and admissibility constraints are fixed at the substrate level so that subsequent causal or normative extensions may diverge without retroactive reinterpretation. Here, "axiomatization" is used in a structural sense: it fixes identity conditions and admissible relations as substrate invariants, while leaving causal and normative commitments to be introduced by extensions.

6.7 Grounding the Construction in the Running Cases

The six-regime construction is not an abstract partition. Each case introduced in Section 1 exercises the full structure and illustrates what stable reference under persistent disagreement concretely requires.

6.7.1 Multi-jurisdiction regulatory reporting.

A corporate entity is represented as an obligation-bearing referent (OBL) that persists across jurisdictional differences in how corporate personhood or subsidiary liability is defined. The reporting statute under which it operates is represented as an authority-bearing structure (NOR) that persists across amendments, contested interpretations, and delayed enforcement. Each filing, inspection, or transaction is represented as a time-indexed occurrence (OCC) individuated by when and where it happened, independently of how any jurisdiction evaluates it. The jurisdiction itself is represented as an applicability context (CTX) that scopes which obligations apply to which entities without collapsing into either the statute or the filer. Performance and compliance indicators are represented as descriptive records (REC) that remain comparable across jurisdictions without carrying causal or normative commitments. Any entity that is regulated but does not itself bear obligations, such as a regulated substance or a monitored facility, is represented as an acted-upon referent (ENR). The six-regime construction allows two jurisdictions to refer to the same corporation, the same statute, and the same

filing while maintaining incompatible evaluations of compliance, without either forcing interpretive convergence or losing the shared referential ground required for audit comparison.

6.7.2 Legislative history tracking.

A statute is represented as an authority-bearing structure (NOR) that remains the same normative referent across the legislative acts that enacted it (OCC), the amendments that modified it (OCC), and the administrative decisions that applied it (OCC). The legislature and the agencies that implement the statute are represented as obligation-bearing referents (OBL) whose identity does not depend on any particular enactment event. Jurisdictions and conditional scopes such as the geographic or subject-matter boundaries of the statute’s application, are represented as applicability contexts (CTX) that may be nested, overlapping, or contested without destabilizing reference to the statute itself. Records of legislative proceedings, judicial interpretations, and compliance histories are represented as descriptive indicators (REC) that characterize outcomes without asserting which interpretation of the statute is correct. The construction allows legal historians, policy analysts, and regulators working under incompatible interpretive frameworks to refer to the same statutory authority and the same sequence of enactment events while disagreeing entirely on what the statute means or which application was legitimate.

6.7.3 Cross-institutional provenance.

A research or administrative record is represented as an acted-upon referent (ENR) whose identity and persistence conditions are invariant under the classificatory schemes of any receiving institution. The institution that produced the record and the institutions that subsequently hold or analyze it are represented as obligation-bearing referents (OBL) that remain identifiable as custodians across transfers, mergers, and re-organizations. Policies governing

data use, retention, or sharing are represented as authority-bearing structures (NOR) that persist independently of whether they have been enforced or contested in any particular transfer. Each transfer, annotation event, or re-analysis is represented as a time-indexed occurrence (OCC) individuated by its provenance, independently of how institutions classify the action. Institutional scope conditions, such as which data a policy governs or which entities fall under a data-sharing agreement, are represented as applicability contexts (CTX). Derived indicators, quality metrics, and outcome summaries attached to the record are represented as descriptive records (REC) that remain comparable across institutions whose causal and normative frameworks differ. The construction preserves the provenance chain as a stable referential structure even when no two institutions agree on how the record should be classified, evaluated, or interpreted.

7 Related Work

This paper is situated at the intersection of formal ontology, knowledge representation, and the study of classification under disagreement. Rather than surveying ontology engineering broadly, this section situates the present result relative to prior work on classification under disagreement, identity and roles in formal ontology, and upper-ontology design.

7.1 Classification, Disagreement, and Neutrality

A substantial body of work has shown that classification systems embed institutional, social, and political commitments rather than functioning as neutral descriptive devices (Bowker and Star 1999; Haslanger 2012). In scientific and policy contexts, disagreement is often persistent rather than transient, reflecting incompatible evaluative frameworks rather than resolvable empirical uncertainty (Longino 1990).

These observations motivate the need for representations that remain usable

without assuming interpretive convergence. The present work adopts this motivation but addresses a distinct question: what minimal structural commitments are unavoidable when disagreement is treated as persistent and interpretive commitments are externalized rather than negotiated?

7.2 Identity, Roles, and Ontological Methodology

Formal ontology has long emphasized the role of identity conditions and persistence criteria in category formation. The OntoClean methodology (Guarino and Welty 2002) clarifies why roles are anti-rigid and unsuitable as identity-defining categories (Guarino 1998), and subsequent work on social and institutional entities has further developed these distinctions (Masolo et al. 2003).

This paper builds on these insights but applies them at the substrate level. The present analysis operates at a different level than role-based modeling, deriving substrate-level identity-and-persistence regimes from neutrality requirements rather than from domain classifications.

7.3 Upper Ontologies

Upper ontologies such as BFO (Smith and Ceusters 2015), DOLCE (Gangemi et al. 2002), and UFO (Guizzardi 2005) provide rich accounts of dependence, endurants and perdurants, and social objects. They differ substantially in commitments and optimization objectives, and are not designed primarily as neutral substrates for interoperability under persistent disagreement.

The present work does not compare expressivity or advocate alignment with any particular upper ontology. Instead, it derives lower-bound constraints that apply independently of such frameworks, conditional on the neutrality and stability requirements adopted here. The regime structure derived here can be situated with respect to two widely-used frameworks to clarify where under-constraints arise not as defects in those frameworks but as points

where implementations built on them would need to introduce additional commitments to satisfy the neutrality and stability requirements assumed here.

BFO (Smith and Ceusters 2015) organizes entities under the continuant/occurrent distinction: continuants persist through time while occurrents unfold across it (Arp et al. 2015b). OCC (time-indexed occurrences) maps cleanly onto BFO occurrents. ENR through NOR map onto independent and generically dependent continuants, but BFO does not enforce the regime-level separation between them: obligation-bearing entities (OBL), authority-bearing structures (NOR), and descriptive indicators (REC) are all candidates for Generically Dependent Continuant, leaving their stability-critical distinctions to the implementer. CTX (applicability and scope) has no first-class BFO home; non-spatial applicability contexts such as normative jurisdiction or conditional scope are represented as GDC, effectively conflating CTX with NOR under a BFO-based implementation. Under persistent disagreement, these conflations require hidden discriminators to recover regime-level distinctions, introducing implicit structure of the kind prohibited by the neutrality constraint.

DOLCE (Gangemi et al. 2002) introduces social objects as first-class entities and provides a richer treatment of institutional and normative structure than BFO (Masolo et al. 2003). This makes DOLCE a better fit for OBL and NOR as distinct categories. However, the NOR/REC conflation persists: both normative structures and descriptive indicators are candidates for social object or quality, and the distinction between authority-as-referent and descriptive-record-as-referent is not enforced at the type level. DOLCE’s region-based account of scope is spatiotemporal, leaving normative applicability contexts (CTX) without a structural home distinct from NOR or REC, and the OBL/NOR boundary (between obligation-bearer and normative structure) is mediated by participation relations rather than enforced by regime membership (Masolo et al. 2003). As with BFO, implementations resolving these under-constraints under persistent disagreement silently introduce hidden regimes.

These observations do not constitute objections to BFO or DOLCE as general-purpose ontological frameworks. They locate a specific point where the optimization objectives of those frameworks (expressiveness, metaphysical coverage, alignment with scientific practice) diverge from the optimization objective assumed here, and where that divergence has structural consequences for neutral interoperability under persistent disagreement.

When constrained to enforce conservative extension over identity and admissibility conditions, such frameworks can realize the regime structure derived here. The contribution of this paper is to identify the structural constraints required when neutrality under persistent disagreement is taken as the primary objective.

The present result is therefore not a competing upper ontology, but a constraint on any ontology intended to function as a neutral, stability-preserving substrate under persistent disagreement.

7.4 Relation to Common Logic and Ontology Exchange

The neutral substrate and derived regimes can be expressed in ISO/IEC 24707 Common Logic (ISO/IEC 2018) as a conservative theory, supporting interpreter-independent ontology exchange without reliance on description-logic restrictions or fixed signatures. Here, conservative means that extensions add structure without altering the consequences derivable about substrate-level terms (Hodges 1997), ensuring that incompatible interpretive frameworks do not retroactively change the meaning of shared references. This enables compatibility with ontology exchange workflows widely used in applied ontology (Kutz et al. 2010) and digital twin infrastructures (Grieves and Vickers 2016), where multiple interpretive layers must interoperate over stable referents.

7.5 Causality, Explanation, and Infrastructure

Separating descriptive structure from causal or explanatory claims is a well-established principle in causal analysis (Pearl 2009). Similarly, research on large-scale information infrastructures emphasizes the importance of long-term stability and interpretive restraint across institutional boundaries (Edwards 2011).

These perspectives align with the pre-causal and pre-normative constraints assumed in this paper. The ontology analyzed here is not an explanatory theory, but an infrastructural substrate intended to support explanation, evaluation, and policy analysis without embedding them.

7.5.1 Relation to FAIR Data Principles

The FAIR principles promote data reuse across communities and over time, but do not specify the ontological constraints required for interoperability under persistent disagreement. In practice, reuse is often pursued through negotiated vocabularies or shared interpretations, which presuppose forms of convergence that may not be available in legal, political, or analytic settings.

The results of this paper clarify a structural precondition for FAIR-aligned reuse in pluralistic or institutionally heterogeneous contexts: stable reference must be supported independently of causal or normative agreement. The identity-and-persistence regimes derived here therefore constrain the ontological substrates upon which FAIR-compliant systems implicitly rely when reuse occurs under persistent disagreement.

8 Conclusion

This paper has addressed a structural question that arises when ontologies are required to function as neutral substrates under persistent disagreement. Under explicit constraints of neutrality, stability under incompatible extension,

and the exclusion of hidden structural discriminations, we have shown that a lower bound exists on the individuation structure required to support accountability. Ontologies with fewer than six distinct identity-and-persistence regimes cannot satisfy the stated neutrality and stability requirements under the assumptions fixed in Section 2. A construction with exactly these six regimes suffices without introducing hidden or excess structure.

The sufficiency construction shows that this boundary is achievable: a substrate realizing exactly six identity-and-persistence regimes satisfies all stated requirements without hidden or excess structure. The result establishes not only that certain distinctions are unavoidable, but that the minimal structure required to support accountability under persistent disagreement is determinate.

The analysis shows that accountability under persistent disagreement requires a cross-cutting stability constraint governing identity and persistence across all represented entities, together with explicit representation of obligation-bearing entities, normative structures, time-indexed occurrences, applicability contexts, and descriptive records, each governed by distinct identity conditions.

The result is conditional rather than prescriptive. It does not claim that the identified structure is optimal for all applications, nor does it assess the adequacy of alternative modeling strategies for other objectives. It establishes only that, under a particular and explicitly stated optimization objective, certain distinctions are structurally unavoidable.

This work clarifies the design space for neutral substrates under persistent disagreement. Under the stated constraints, it shows that the lower bound on explicit ontological structure is determined by structural requirements rather than by modeling preference. Future work may examine how alternative optimization objectives yield different structural bounds, or analyze how the six-regime construction behaves when stability or neutrality assumptions are systematically weakened. Such analyses may help clarify how different interoperability objectives generate different minimal ontological structures.

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The author reviewed all suggestions and takes full responsibility for the content of this work.

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Author Contributions

The author was solely responsible for the conception, analysis, and writing of this manuscript.

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Appendix A. Complete Pairwise Collapse Enumeration

All $\binom{6}{2} = 15$ unordered pairs among the six regimes are enumerated below and classified by primary failure mode under the admissibility criterion of Section 4.3. Each pair appears exactly once in the enumeration.

Identity Instability

Six pairs cannot be collapsed without violating extension-invariant identity.

- OBL/NOR (incompatible persistence criteria: obligation-bearing entities persist across delegation and compliance, whereas normative structures persist across enactment, amendment, and repeal);
- OBL/OCC (incompatible persistence criteria: obligation-bearing entities are enduring referents, whereas occurrences are individuated by temporal realization and do not persist through modification);
- OCC/ENR (incompatible persistence criteria: occurrences are time-indexed realizations, whereas enduring referents persist independently of temporal realization);
- OCC/NOR (incompatible persistence criteria: occurrences are individuated by realization events, whereas normative structures persist across enactment, amendment, and repeal);
- OCC/CTX (incompatible persistence criteria: occurrences are time-indexed realizations, whereas applicability contexts persist as enduring scoping referents);
- OCC/REC (incompatible persistence criteria: occurrences are individuated by temporal realization, whereas descriptive records persist as enduring referents whose identity is not determined by any single realization event).

Category Error

Six pairs cannot be collapsed without producing a category error under the neutrality constraint.

- OBL/ENR (normative attribution relations appropriate to obligation-bearing entities would become structurally admissible for all enduring referents, misclassifying non-normative entities as potential obligation-bearers);
- OBL/CTX (obligation-bearing entities admit normative attribution relations, whereas applicability contexts admit scoping relations, yielding incompatible admissible-relation classes);
- NOR/CTX (normative structures admit modification relations such as enactment and amendment, whereas applicability contexts admit scoping relations such as nesting and refinement);
- NOR/ENR (normative grounding relations appropriate to normative structures would misclassify purely enduring referents as sources of authority);
- CTX/ENR (scoping relations appropriate to applicability contexts would misclassify ordinary enduring referents as scope-defining entities);
- CTX/REC (scoping relations appropriate to applicability contexts would misclassify descriptive records as scope-defining referents).

Hidden Regime Reintroduction

Three pairs cannot be collapsed without reintroducing the original distinction through hidden discriminators.

- NOR/REC (normative grounding relations would require an *isNormative* discriminator to prevent descriptive records from functioning as normative authorities);
- OBL/REC (normative attribution relations would require a discrimina-

tor such as *isObligationBearer* to prevent descriptive records from being treated as obligation-bearing entities, reintroducing the distinction at the extension layer);

- ENR/REC (measurement targets and descriptive records would require an *isDescriptive* flag to distinguish descriptive records from the entities they describe).

Appendix B. Glossary of Terms

This glossary defines key terms used in the paper.

Applicability Context

An enduring referent that delimits the scope under which a normative structure applies, including spatial, jurisdictional, or conditional applicability. Applicability contexts persist independently of specific occurrences and must support nesting and overlap.

Hidden Regime

A distinction relevant to identity or persistence that is not represented as an explicit ontological kind, but is instead encoded via roles, flags, contextual predicates, or interpretive conventions supplied by extension layers. Hidden regimes are excluded under the stability optimization adopted here.

Identity-and-Persistence Regime

A structured set of identity and persistence conditions that determine how a class of entities can be stably referenced over time within an ontology. An identity-and-persistence regime specifies what counts as the same entity, what changes are compatible with continued identity, and how references to that entity remain stable across datasets, jurisdictions, or observational contexts. In a neutral substrate, such regimes provide the identity and persistence structure upon which interpretive frameworks operate. In this paper, such regimes are used to characterize the minimal structure required to support stable reference under the neutrality constraint.

Individuation

The specification of the identity and persistence conditions by which entities are distinguished and tracked as the same or different across time and contexts.

Interpretive Extension

Causal or normative structure introduced by extensions that provides explanation or judgment without altering substrate-level identity or admissibility conditions.

Lower-Bound Result

A necessity claim establishing that no representation with fewer than a specified number of identity-and-persistence regimes can satisfy the stated requirements under the adopted optimization criterion.

Neutral Substrate

Causal or normative structure introduced by extensions that provides explanation or judgment without altering substrate-level identity or admissibility conditions. A neutral substrate excludes causal and normative commitments from the foundational layer, supporting pluralism and accountability.

Neutrality Constraint

The requirement that the substrate itself remain pre-causal and pre-normative, permitting incompatible interpretations without requiring revision of entity identity or structure.

Optimization Criterion

The objective function fixed for this paper: to minimize explicit ontological structure subject to neutrality, stable reference, and the exclusion of hidden identity regimes. Minimality is defined relative to stability, not syntactic brevity.

Persistent Disagreement

A condition in which interpretive, normative, or analytic incompatibilities are expected to persist over time, rather than converge through negotiation, evidence, or governance.

Stable Reference

The property that entities remain referable as the same entities

across time and across incompatible extensions, without reclassification or reinterpretation of identity criteria.

Sufficiency Construction

An explicit ontological construction that satisfies all stated requirements using exactly the minimal number of identity-and-persistence regimes established by the lower-bound result.

Tight Bound

A result establishing that the same number of regimes is both necessary and sufficient under the stated assumptions and optimization criterion.

Time-Indexed Occurrence

An entity whose identity is inseparable from its temporal realization and provenance, and which does not persist beyond its occurrence.

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